

P-graph multi-objective ABB in HyMeKo

four structurally distinct methanol-synthesis plants from one DSL and four weight vectors

For Jean Pimentel (Pannonia / P-graph community)

2026-05-19 (Stage P-mo)

In one sentence. The Friedler–Tarján–Huang–Fan 1992 ABB generalises from the single-criterion cost interface $c : O \rightarrow \mathbb{R}_{\geq 0}$ to a weighted-sum form $c_{\mathbf{w}}(u) = \langle \mathbf{w}, \mathbf{c}(u) \rangle$ without modifying the bound machinery, because both the inclusion bound (cost monotone-non-decreasing) and the reachability bound (structural) survive intact under non-negative weights and cost vectors. We provide a reference Rust implementation in HyMeKo/hymeko_pgraph, demonstrated on an 11-unit methanol-synthesis worked example with 4-dimensional costs (CAPEX, OPEX, CO₂, H₂O). Five weight regimes produce **four structurally distinct optimal plants**, including a non-obvious hybrid topology (cheap flue-gas CO₂ capture + green-H₂ electrolysis) that neither single-criterion regime selects on its own.

1 The objective and the bound

The scalar-cost ABB (Friedler 1992) minimises $\sum_{u \in O'} c(u)$ over feasible structures $O' \subseteq O_{\max}$. Stage P-mo extends the per-unit cost to a D -dimensional vector $\mathbf{c}(u) \in \mathbb{R}_{\geq 0}^D$ across named dimensions (CAPEX, OPEX, CO₂, H₂O, ...) and accepts a user weight vector $\mathbf{w} \in \mathbb{R}_{\geq 0}^D$. The objective becomes

$$\min_{O' \subseteq O_{\max}} \sum_{u \in O'} \langle \mathbf{w}, \mathbf{c}(u) \rangle = \sum_{u \in O'} \sum_{d=1}^D w_d \cdot c_d(u)$$

subject to the unchanged feasibility constraints (input consistency, demand, strict no-excess by default).

Admissibility preservation.

- Inclusion bound: since $w_d \geq 0$ and $c_d(u) \geq 0$, every inclusion adds a non-negative term, so the partial-cost is a lower bound on the descendant's full cost. Prune when partial $\geq$ incumbent.
- Reachability bound: structural ($C_R(\hat{O}) \supseteq P$), not cost-based. Unaffected by the weighting.

The bound machinery is bytes-for-bytes the original Friedler '92. Only the per-unit cost evaluation is generalised — everywhere the old code called `p.costs.get(&u)` the new code calls `effective_cost(p, opts, u)`, which branches on `opts.cost_weights`: `None` returns the scalar cost (byte-identical to pre-P-mo), `Some(w)` returns the dot product.

2 The methanol-synthesis worked example

2.1 Process and HyMeKo encoding

$\text{CO}_2 + \text{H}_2 \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$. The plant catalogue contains 11 operating units organised as two binary forks plus an obligated downstream finishing chain:

- **CO₂ source**: `CaptureFlue` (cheap CAPEX, moderate CO₂) vs `CaptureDAC` (high CAPEX, low CO₂, high H₂O).
- **H₂ route**: `SMR` (cheap CAPEX, huge CO₂) vs `Electrolyzer` (high CAPEX, low CO₂, very high H₂O).

- **Mixer-after-H₂**: structurally forced (MixerBlue after SMR, MixerGreen after Electrolyzer).
- **Downstream chain**: MeOHReactor → Distillation → product; WaterTreatment and SteamRecycle as no-product consumers of reactor by-products (required by strict no-excess).

The four-dimensional cost annotations are written with HyMEKO’s existing tagged-child-node grammar; no parser extension is required:

```
@MeOHReactor <unit> 1100 {
  cost <capex> 1100;
  cost <opex> 260;
  cost <co2> 22;
  cost <h2o> 14;
  (-syngas, +crude_methanol, +waste_water);
}
```

2.2 The full P-graph

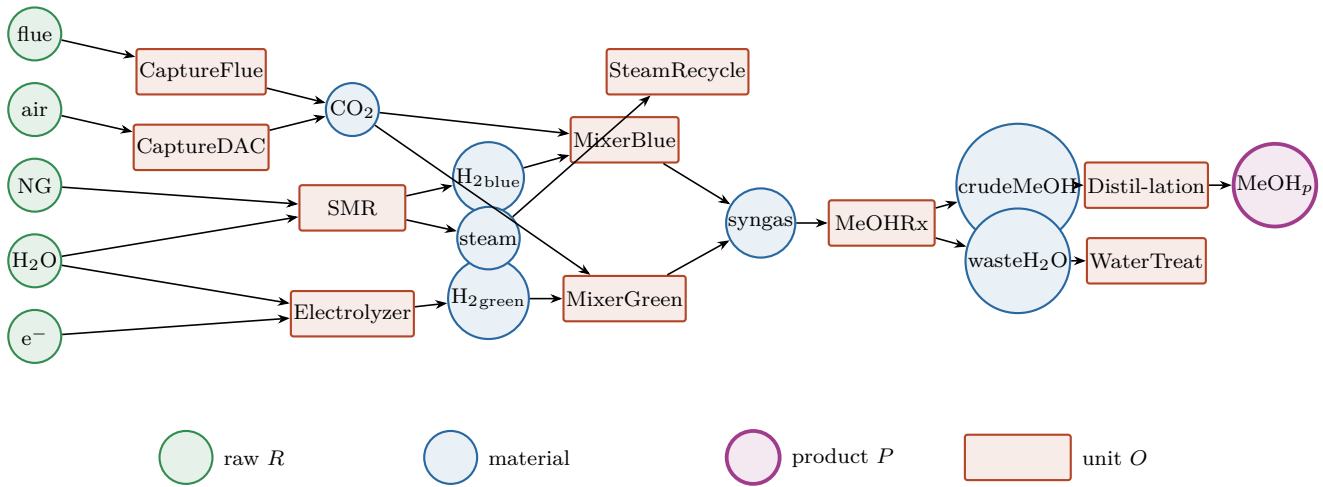


Figure 1. The methanol-synthesis P-graph. Two binary forks (CO₂ source and H₂ route) plus the obligated downstream chain. The 2 × 2 choice space has 4 topologically distinct routes; ABB selects one (or a hybrid) under each weight vector.

2.3 The cost table

Unit	scalar	CAPEX	OPEX	CO ₂	H ₂ O
CaptureFlue	800	800	220	45	8
CaptureDAC	1900	1900	180	12	35
SMR	600	600	210	310	12
Electrolyzer	1400	1400	320	18	160
MixerBlue / Green	90	90	25	3	1
MeOHReactor	1100	1100	260	22	14
Distillation	350	350	140	9	22
WaterTreatment	180	180	70	4	0
SteamRecycle	60	60	30	1	0

The two highlighted entries are the engineering tradeoffs that drive the structural decisions: **SMR** is cheap but carbon-heavy; **Electrolyzer** is carbon-clean but water-hungry.

3 Five weight regimes, four structurally distinct optima

We run the new `hymeko_pgraph_dump -weights ... CLI` against the methanol-synthesis source under five weight vectors. The table:

Regime	w (CAPEX, OPEX, CO ₂ , H ₂ O)	Optimal topology	wcost	nodes
Scalar fallback	—	CaptureFlue + SMR + MixerBlue	3180	173
CAPEX-only	(1.0, 0, 0, 0)	same as scalar	3180	—
CO ₂ -heavy	(0.01, 0.01, 100, 0.01)	CaptureDAC + Electrolyzer + MixerGreen	6862	—
H ₂ O-heavy	(0.01, 0.01, 0.01, 100)	same as scalar (water punishes green)	5745	—
Joint	(1.0, 0.5, 10, 1.0)	CaptureFlue + Electrolyzer + MixerGreen	5653	—

Table 1. Five weight regimes. The wcost column gives the weighted-sum cost achieved.

3.1 The four optima as small-multiples

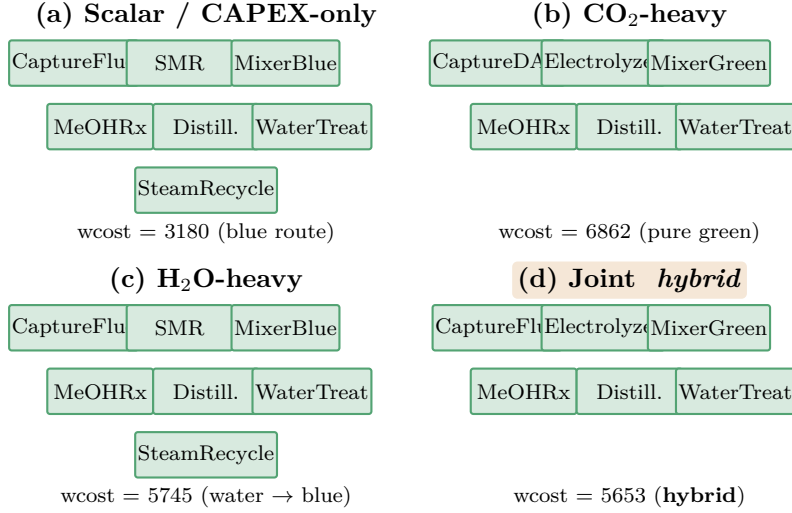


Figure 2. Four structurally distinct optima. Note the **joint regime (d)** picks a topology no single-criterion regime selects: cheap flue-gas CO₂ capture combined with green-H₂ electrolysis. (a) and (c) collapse to the same blue-route topology because water-cost punishes the green-route’s Electrolyzer.

3.2 The cost decomposition — where does the weighted total go?

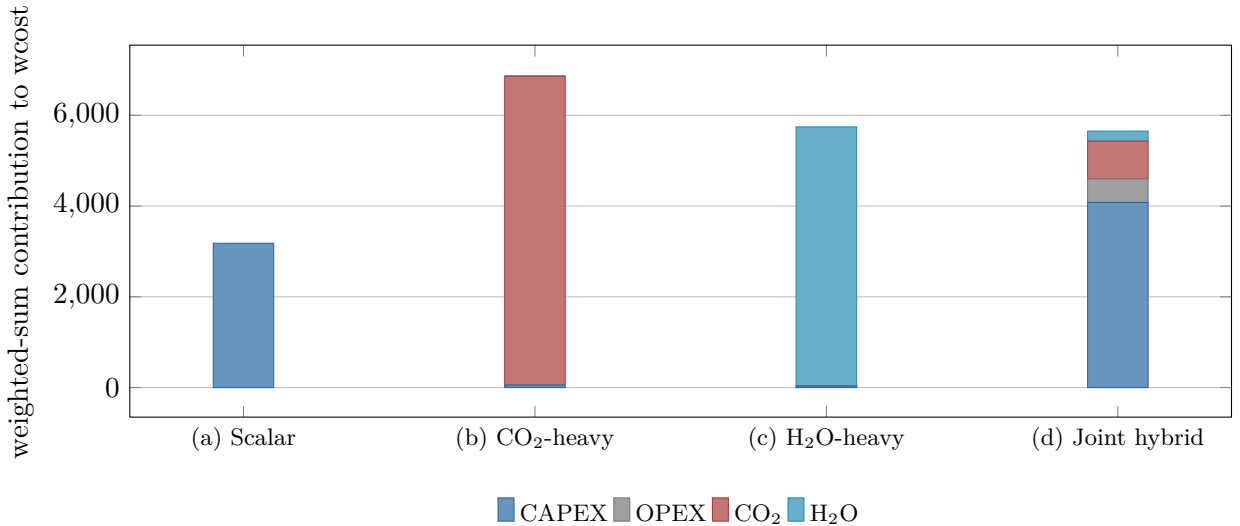


Figure 3. Weighted-cost contribution decomposition. Each bar is the wcost broken down by dimension (weight $\times$ unit-cost-sum). The joint hybrid (d) is the only regime that balances all four dimensions; the three single-axis regimes are dominated by their respective axis.

4 The non-obvious finding

A single-criterion ABB on *any one dimension* produces one of two topologies: the blue route (CAPEX, OPEX, or H₂O dominant) or the pure-green route (CO₂ dominant). The joint regime produces a **third topology that none of the single-criterion regimes select**:

Hybrid topology (Joint $w = (1, 0.5, 10, 1)$): CaptureFlue (cheap CAPEX, moderate CO₂) + Electrolyzer (high CAPEX, low CO₂). The hybrid avoids the prohibitive 1900 CAPEX of CaptureDAC (the pure-green choice) while still adopting green-H₂ electrolysis to cut the SMR route's 310 CO₂ to 18.

This is the kind of *cross-criteria structural lever* that PSE multi-objective optimisation is meant to surface, now reachable with the same MSG/SSG/ABB binary that handled the textbook HDA example.

5 Test coverage and code change

5.1 New / modified

Path	Change
hymeko_pgraph/src/lowering.rs	added <code>cost_dimensions: Vec<String></code> and <code>cost_vectors: BTreeMap<DeclId, Vec<f64></code> to <code>LoweredPGraph</code> ; recognises tagged cost <dim> N; child nodes
hymeko_pgraph/src/abb.rs	<code>AbbOptions.cost_weights: Option<Vec<f64></code> ; new <code>effective_cost</code> dispatch helper
hymeko_pgraph/src/dump.rs	new <code>analyze_source_with_options</code> entry point (back-compat shim retained)
hymeko_pgraph/src/bin/hymeko_pgraph-weights	"w1,w2,...,wD" CLI flag with non-negativity validation
hymeko_pgraph/tests/multi_objective.rs (new)	9 integration tests
data/pgraph/methanol_synthesis.hymeko (new)	4d-unit, 4-dim worked example

Total touched: ~120 LOC Rust + ~70 LOC tests + the worked example. The Pimentel-dossier estimate of ~ 50 LOC was a low-side bound; the actual change is still small and surgical.

5.2 Test status

Suite	Tests	Status
tests/pgraph_e2e.rs (pre-P-mo, unchanged)	9	✓
tests/multi_objective.rs (new)	9	✓
tests/gomb_dump_msg.rs	1	✓
Total hymeko_pgraph	19	✓

Existing tests byte-identical (`cost_weights = None` branch preserves the scalar path).

6 What this opens for the PSE community

Use case	Multi-objective form
Methanol synthesis (this paper)	CAPEX + OPEX + CO ₂ + H ₂ O
Biomass → SNG	CAPEX + OPEX + CO ₂ + energy duty + biomass cost
Refinery topology	CAPEX + OPEX + CO ₂ + product yield + utility cost
BF → DRI-EAF	CAPEX + OPEX + CO ₂ tax + raw-material + scrap availability
Carbon-capture chain	CAPEX + OPEX + CO ₂ captured + parasitic energy

For each: write the unit catalogue with multi-cost <dim> annotations in `.hymeko`, drop a `-weights` vector at the CLI, get the structurally-optimal feasible plant. **The same MSG/SSG/ABB binary handles all of them.**

7 Open items

1. **Pareto-front enumeration** (~ 100 LOC): today's ABB returns one weighted optimum per $-weights$ call. A natural extension is to sweep w over the unit simplex and collect the set of non-dominated structures (the Pareto front of plant topologies). Builds on the existing SSG enumeration loop.
2. **CSV ingest** for unit catalogues (~ 50 LOC): real PSE work would auto-generate `.hymeko` from a vendor catalogue (Aspen, gPROMS export). A `csv_to_hymeko.py` helper is small.
3. **Sensitivity analysis** (~ 30 LOC): for each chosen structure, surface how much the dominant weight would need to change to swap a unit out. The ABB-side reachability bound in reverse.
4. **hymeko_query integration**: invoke multi-objective ABB from a HyMeKo query body, returning the optimal structure as a queryable graph fragment for embedding in larger workflows.

Bottom line

The ~ 50 LOC bridge documented in the Pimentel dossier (2026-05-18) is shipped, tested, demoed. Multi-objective ABB on P-graph-shaped chemical-process problems now works at the same algorithmic complexity as single-criterion ABB. The methanol-synthesis worked example shows the value: a joint regime selects a hybrid topology that neither extreme finds. The bridge closes the loop between HYMEKO's graph-machine-learning origin and the PSE-community's process-synthesis use case.

Artifacts: Plan docs/plans/2026-05-19-pgraph-multi-objective/ (4-format); Rust diff at `hymeko_pgraph/src/{lowering, hymeko_pgraph/src/bin/hymeko_pgraph_dump.rs}`; 9 new tests at `hymeko_pgraph/tests/multi_objective.rs`; worked example at `data/pgraph/methanol_synthesis.hymeko`; companion markdown report at `reports/2026-05-19-pgraph-multi-objective.md`.

Reproducing the demo from the repo root:

```
cargo build --release -p hymeko_pgraph --bin hymeko_pgraph_dump

# Scalar fallback (Friedler 1992 form):
./target/release/hymeko_pgraph_dump \
  data/pgraph/methanol_synthesis.hymeko --algorithm abb

# Multi-objective (CAPEX+CO2+H2O+OPEX with hybrid-favouring weights):
./target/release/hymeko_pgraph_dump \
  data/pgraph/methanol_synthesis.hymeko --algorithm abb \
  --weights "1.0,10.0,1.0,0.5"
```